Cambridge Assessment International Education

Cambridge IGCSE™

PHYSICS **0625/42**

Paper 4 Extended Theory **May/June 2020**

MARK SCHEME

Maximum Mark: 80

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **9** printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

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5 <u>'List rule' guidance</u>

For questions that require ***n*** responses (e.g. State **two** reasons …):

- The response should be read as continuous prose, even when numbered answer spaces are provided
- Any response marked *ignore* in the mark scheme should not count towards ***n***
- Incorrect responses should not be awarded credit but will still count towards ***n***
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 <u>Calculation specific guidance</u>

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 <u>Guidance for chemical equations</u>

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)(i)	s = vt in any form OR (s =) vt OR relates distance to area (under graph)	C1
	any **one** of: 5 × 20 / 60 OR 40 × 20 / 60 OR 6 × 22 / 60	C1
	(s = 1.667 + 13.333 + 2.2 =) 17 km	A1
1(a)(ii)	average speed = candidate's (i) / time	C1
	(average speed = 17 × 60 / 74 =) 14 km / h	A1
1(b)	gradient	B1
	(gradient =) change of speed / time	B1
1(c)	0	B1
	(constant) gradient = 0 OR speed constant	B1

Question	Answer	Marks
2(a)	impulse OR Δp = m(v – u) in any form	C1
	(impulse =) 750 000 (84 – 42)	C1
	(impulse =) 3.2×10^7 N s or m kg / s	A1
2(b)	Ft = impulse OR Δp in any form OR (F =) (impulse OR Δp) / t	C1
	(F = 3.2 x 10^7 / 80 =) 3.9×10^5 N	A1
2(c)	reduces drag / air resistance (experienced by the train) / more streamlined	B1
2(d)	less drag / air resistance (at slower speeds)	B1
2(e)	(maximum) friction (force) between rails and train reduced / train may slide	B1

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Question	Answer	Marks
3(a)	E = mcΔT in any form OR (E =) mcΔT	C1
	efficiency = (energy) output / (energy) input in any form	C1
	15 × 4200 × ΔT = 5000 × 3600 × 0.2	C1
	(ΔT = 5000 × 3600 × 0.2 / 15 × 4200 =) 57 °C	A1
3(b)	e.g. renewable OR no air pollution OR low running costs OR no named polluting gas OR no greenhouse effect	M1
	explanation that follows from advantage stated	A1
	e.g. expensive to install OR not available at night OR visual pollution OR needs a suitable (roof) space	M1
	explanation that follows from disadvantage stated	A1

Question	Answer	Marks
4(a)	(place) in melting ice	B1
	when bead has stopped moving OR owtte mark as lower fixed point / 0 °C	B1
	(place) in steam (above boiling water)	B1
	when bead has stopped moving OR owtte mark as upper fixed point / 100 °C	B1
4(b)	bead would not be liquid owtte	B1
4(c)(i)	thinner bore / tube OR smaller bulb OR use liquid which expands more (per unit change in temperature)	B1
4(c)(ii)	longer tube OR larger (volume) glass bulb OR use liquid which expands less (per unit change in temperature)	B1
4(d)	expands uniformly (with temperature) OR same distance between all degree intervals	B1

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Question	Answer	Marks
5(a)	three wavefronts parallel to each other AND same angles of reflection and incidence both by eye	B1
	two wavelengths same as original wavelength by eye	B1
	three reflected waves meet incident waves at barrier	B1
5(b)	$v = f\lambda$ in any form OR (f =) v/λ	C1
	OR (f =) 1.2 / 0.36	C1
	(f =) 3.3 Hz	A1
5(c)	sound OR ultrasound	B1
	compressions	B1

Question	Answer	Marks
6(a)	incident ray travels straight on at first face of prism 1	B1
	ray reflected through 90° at sloping face of prism 1 continues vertically downwards to sloping face of prism 2	B1
	ray reflected through 90° at sloping face and leaves box horizontally	B1
6(b)	n = 1 / sin C in any form OR (n =) 1 / sin C	B1
	{(n=) 1 / sin 45 OR (n=) 1 / 0.707} AND (n=) 1.41	B1

Question	Answer	Marks
7(a)(i)	primary coil wound round iron AND (labelled primary or coil OR connected to labelled supply / 12 V)	B1
	secondary coil wound round iron AND (labelled secondary OR connected to labelled output / 2 V)	B1
	significantly more coils around primary	B1

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Question	Answer	Marks
7(a)(ii)	two from three • links magnetic fields of coils / primary and secondary • stronger magnetic field in secondary • better induction owtte	**B2**
7(a)(iii)	$V_1I_1 = V_2I_2$ in any form OR (I_1 =) V_2I_2 / V_1 OR (I_1 =) $2 \times 0.10 / 12$	**C1**
	(I_1 =) 0.017 A	**A1**
7(b)	metal case earthed	**B1**
	in case wire falls off / insulation fails / live(wire) touches case	**B1**

Question	Answer	Marks
8(a)	two circuit symbols correct	**B1**
	three circuit symbols correct	**B1**
	symbol for cell, battery or power supply AND two other circuit symbols in series	**B1**
	LED correct way round	**B1**
8(b)	R = V / I in any form OR (R =) V / I	**C1**
	(R = 3.1 / 0.030 =) 100 Ω	**A1**
8(c)(i)	uses 10.5 = 2.1 + V across heater	**C1**
	(R = 8.4 / 1.5 =) 5.6 Ω	**A1**
8(c)(ii)	P = VI in any form OR (P =) VI	**C1**
	(P = 8.4 × 1.5 =) 12.6 W	**A1**

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Question	Answer	Marks
9(a)	output 1, 1, 1, 0	**B1**
9(b)(i)	AND	**B1**
9(b)(ii)	first two lines of E 0,1	**B1**
	last two lines of E 1,1	**B1**
9(c)	Q = It in any form OR (Q =) It	**C1**
	$(Q = 3 \times 60) = 180$ (C)	**C1**
	$(n = 3 \times 60 / 1.6 \times 10^{-19}) = 1.1 \times 10^{21}$	**C1**

Question	Answer	Marks
10(a)	(beam) narrow OR straight OR in one direction owtte	**B1**
	radiation in other directions absorbed	**B1**
10(b)(i)	out of page / towards viewer	**B1**
	equiv. to current in direction of beam	**B1**
10(b)(ii)	opposite to (i)	**B1**
	equiv. to current in opposite direction to beam OR LH rule mentioned or described	**B1**
10(b)(iii)	none	**B1**
	(γ) uncharged OR not equivalent to current	**B1**

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